

Pearls AQI Predictor: Exploratory Data Analysis of Air Quality in Karachi

Milestone 4 Technical Report

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Abstract—This report presents the Exploratory Data Analysis (EDA) phase of the Pearls AQI Predictor, a serverless machine learning system designed to forecast the Air Quality Index (AQI) of Karachi three days in advance. Six months of hourly historical data (4,344 records) were collected from the Open-Meteo Weather and Air Quality APIs, stored in the Hopworks Feature Store, and analyzed to uncover temporal patterns and feature relationships. The analysis reveals that Karachi’s air quality is moderate on average (mean AQI 90), with pronounced seasonal and diurnal variation. PM_{2.5} was identified as the dominant predictor of AQI, while meteorological variables such as wind speed and temperature show meaningful negative correlations. These insights directly inform the feature selection and modeling strategy for the subsequent training pipeline.

Index Terms—air quality index, exploratory data analysis, feature store, Hopworks, PM_{2.5}, machine learning

I. INTRODUCTION

Air pollution is a persistent public health challenge in Karachi, where seasonal and meteorological factors drive frequent fluctuations in the Air Quality Index (AQI). The objective of the Pearls AQI Predictor is to build an end-to-end serverless system that forecasts AQI three days ahead. Before any predictive model can be trained, it is essential to understand the underlying structure of the data through Exploratory Data Analysis (EDA).

This milestone focuses on retrieving the historical dataset from the Hopworks Feature Store, validating its integrity, and analyzing the temporal and statistical patterns that characterize Karachi’s air quality. The findings establish a foundation for informed feature selection in the modeling phase.

II. DATA COLLECTION AND STORAGE

The dataset comprises 4,344 hourly records spanning approximately six months, from late November 2025 to late May 2026. Two complementary data sources were used:

- **Weather data** was obtained from the Open-Meteo Archive API, providing temperature, relative humidity, wind speed, and surface pressure.
- **Pollution data** was obtained from the Open-Meteo Air Quality API, providing PM_{2.5}, PM₁₀, nitrogen dioxide (NO₂), ozone (O₃), sulphur dioxide (SO₂), and carbon monoxide (CO).

The AQI value was computed from the PM_{2.5} concentration using the United States Environmental Protection Agency (US EPA) breakpoint formula:

$$AQI = \frac{I_{hi} - I_{lo}}{C_{hi} - C_{lo}}(C - C_{lo}) + I_{lo} \quad (1)$$

where C is the measured PM_{2.5} concentration, C_{lo} and C_{hi} are the concentration breakpoints surrounding C , and I_{lo} and I_{hi} are the corresponding AQI breakpoints.

All engineered features, including time-based attributes (hour, day of week, month, weekend indicator) and the categorical AQI health class, were stored in the Hopworks Feature Store as the feature group `aqi_features`. This centralized storage ensures consistency between the analysis, training, and inference stages of the project.

III. METHODOLOGY

The EDA was conducted in a Jupyter notebook using the conda environment configured for the project. Data was read back from the Hopworks Feature Store to confirm successful storage and to provide the working dataset. The analysis employed the `pandas` library for data manipulation and `matplotlib` and `seaborn` for visualization. The following analyses were performed: descriptive statistics, AQI distribution analysis, temporal analysis (by hour, day, month, and over time), correlation analysis, and pollutant–weather relationship analysis.

IV. RESULTS AND DISCUSSION

A. Descriptive Statistics

The dataset contained no missing values across all 18 features. The AQI ranged from 15 to 172 with a mean of 90.0 and a standard deviation of 28.7, indicating substantial variation suitable for predictive modeling. Table I summarizes the key statistics.

B. AQI Distribution

The distribution of AQI categories showed that Karachi’s air quality is predominantly moderate (68.1% of observations), followed by unhealthy for sensitive groups (23.0%), unhealthy (6.5%), and good (2.4%). The histogram of AQI values exhibited a primary concentration around the value of 80, with a secondary cluster near 150 corresponding to pollution spikes.

TABLE I
SUMMARY STATISTICS OF AQI

Statistic	Value
Total records	4,344
Mean AQI	90.0
Standard deviation	28.7
Minimum AQI	15.0
Maximum AQI	172.0
Time unhealthy (AQI > 150)	6.5%

C. Temporal Patterns

Several temporal patterns emerged from the analysis:

- **Diurnal pattern:** Air quality was worst in the early morning, peaking around 3:00–4:00 AM (mean AQI approximately 103), and cleanest around noon (mean AQI approximately 79). This pattern is consistent with nighttime temperature inversions that trap pollutants near the ground.
- **Weekly pattern:** AQI remained relatively constant across all days of the week (88–92), suggesting that pollution is driven more by meteorological and industrial factors than by weekday traffic cycles.
- **Monthly pattern:** November was the most polluted month (mean AQI approximately 108), followed by December and January, while the spring months of March through May were considerably cleaner (76–80). This confirms a strong seasonal winter pollution effect.

D. Correlation Analysis

The correlation heatmap revealed that PM_{2.5} is almost perfectly correlated with AQI (0.99), confirming its role as the dominant driver. Carbon monoxide (0.57), PM₁₀ (0.58), and nitrogen dioxide (0.55) showed moderate positive correlations. Wind speed (−0.34) and temperature (−0.33) were negatively correlated with AQI, indicating that windier and warmer conditions are associated with cleaner air through pollutant dispersion.

E. Pollutant and Weather Relationships

Scatter plots confirmed the near-linear relationship between PM_{2.5} and AQI, with a slight slope change near AQI 150 corresponding to a change in the EPA breakpoint segment. Wind speed exhibited a clear downward trend against AQI, reinforcing its dispersive effect, whereas humidity and temperature showed weaker, more scattered relationships.

V. CONCLUSION

The exploratory data analysis confirmed that the collected dataset is complete, varied, and well-suited for predictive modeling. Karachi’s air quality follows clear diurnal and seasonal patterns, with PM_{2.5} as the strongest predictor of AQI and wind speed as a meaningful secondary meteorological factor. These insights will guide feature selection and model design in the subsequent training pipeline milestone, in which regression and deep learning models will be trained and evaluated, with the best-performing model registered in the Hopsworks Model Registry.